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Project Phase 2 Report on

**“ A Swarm Intelligent Framework for Knowledge Transfer Between
Heterogeneous Machine Learning Models in Multi Robot Systems “**

**A dissertation submitted in partial fulfillment of the requirements for the award of the
degree of M.Tech in Computer Science and Engineering**

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CERTIFICATE

This is to certify that the dissertation entitled “**A Swarm Intelligent Framework For Knowledge Transfer Between Heterogeneous Machine Learning Models In Multi Robot Systems**” is a bonafide work carried out by **Mr. S MD Zaheed Hussain** bearing USN **KUB24MCS018** in partial fulfillment of the requirements for the award of the **Master Degree in Computer Science & Engineering** of the **Kishkinda University** during the academic year 2025-2026, under my supervision and guidance. To the best of my knowledge, this work has not been submitted elsewhere for the award of any other degree or diploma.

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EXAMINERS

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, **S MD Zaheed Hussain** bearing USN **KUB24MCS018**, hereby declare that the dissertation entitled “**Swarm Intelligent Framework for Knowledge Transfer Between Heterogeneous Machine Learning Models in Multi Robot Systems**” submitted in partial fulfillment of the requirements for the award of the **Master Degree in Computer Science and Engineering** is my original work carried out under the guidance of **Dr. Rajashree V Biradar**. I further declare that this work has not been submitted, either in part or in full, for the award of any other degree or diploma in any university or institution.

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ABSTRACT

Swarm intelligence and multi-agent systems have emerged as effective paradigms for achieving scalable, decentralized, and collaborative decision-making in distributed environments. However, heterogeneous machine learning models deployed in multi-agent systems often operate independently, leading to redundant inference, increased computational overhead, and limited knowledge reuse. Existing approaches such as federated learning and swarm learning primarily focus on homogeneous architectures and training-level collaboration, making them unsuitable for real-time knowledge exchange among heterogeneous models.

This project presents a swarm-intelligent, model-agnostic framework for knowledge transfer among heterogeneous machine learning models. The proposed system employs three heterogeneous perception agents based on MobileNet, EfficientNet, and ResNet architectures for human detection from video streams. Each agent generates structured knowledge packets containing semantic information such as object labels, confidence scores, feature embeddings, timestamps, and agent identifiers. These packets are shared through a collective memory module implemented using a distributed communication mechanism.

For each video frame, agents store and retrieve relevant knowledge packets from the collective memory, enabling knowledge reuse and reducing redundant processing. The framework further incorporates an adaptation mechanism that allows agents to learn from shared knowledge representations, thereby improving their perception capabilities over time. Experimental evaluation demonstrates enhanced collaborative perception, reduced redundant inference, efficient knowledge reuse, and improved performance after knowledge transfer among heterogeneous models.

The proposed framework validates the feasibility of real-time knowledge sharing in heterogeneous multi-agent systems and provides a foundation for future swarm-intelligent robotic applications, including autonomous navigation, surveillance, distributed sensing, and collaborative AI systems.

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LIST OF ABBREVIATIONS AND ACRONYMS

S NO.	Abbreviation	Full Form
1	CM	Collective Memory
2	HMLM	Heterogeneous Machine Learning Models
3	KT	Knowledge Transfer
4	KP	Knowledge Packet
5	MQTT	Message Queuing Telemetry Transport
6	MAS	Multi-Agent System
7	ROS	Robot Operating System
8	SI	Swarm Intelligence
9	UAV	Unmanned Aerial Vehicle

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and autonomous robotic systems has significantly increased the adoption of multi-agent and multi-robot environments across various application domains. Modern robotic systems are increasingly deployed in surveillance, autonomous navigation, disaster response, industrial automation, and smart city applications, where multiple agents must collaborate to achieve common objectives efficiently [2], [6]. To enable such collaboration, researchers have drawn inspiration from swarm intelligence, a computational paradigm based on the collective behavior of social organisms such as ants, bees, and bird flocks. Swarm-intelligent systems are characterized by decentralization, self-organization, scalability, and fault tolerance, making them highly suitable for dynamic and distributed environments [2], [9]. Recent developments in machine learning and deep learning have enabled autonomous agents to perform complex perception tasks such as object detection, image classification, scene understanding, and decision-making with high accuracy [5], [11], [12]. However, due to variations in hardware resources, energy constraints, and deployment requirements, different agents often utilize different machine learning architectures. As a result, modern multi-agent systems are increasingly heterogeneous, comprising agents with varying computational capabilities and perception models [6], [7], [8].

Although heterogeneous models provide flexibility and resource efficiency, they also introduce significant challenges in collaboration and knowledge sharing. In most existing systems, agents perform perception independently, even when observing similar environments or objects. This leads to redundant computation, increased inference latency, higher resource consumption, and inefficient utilization of the collective intelligence available within the swarm [3], [5]. Furthermore, lightweight models operating on resource-constrained agents are unable to directly benefit from the knowledge generated by more accurate or computationally powerful models [1], [6]. Several distributed learning paradigms, including federated learning and swarm learning, have been proposed to facilitate collaboration among multiple agents while preserving data privacy [1], [4]. However, these approaches primarily focus on collaborative

model training and typically assume homogeneous model architectures. Similarly, knowledge distillation techniques enable knowledge transfer from larger models to smaller models but generally require retraining and centralized processing, making them unsuitable for real-time collaborative environments [6], [10].

These limitations highlight the need for a decentralized and model-agnostic framework that enables heterogeneous machine learning models to exchange useful perception knowledge during inference. Such a framework can reduce redundant processing, improve resource utilization, and enable the emergence of swarm-level collective intelligence without requiring raw data exchange or model parameter sharing [1], [2], [10]. The proposed work addresses this challenge through a swarm-intelligent knowledge transfer framework in which autonomous agents share structured knowledge packets and collaboratively improve perception performance using a collective memory mechanism.

1.2 PROBLEM STATEMENT

To design and develop a decentralized swarm-intelligent framework for model-agnostic knowledge transfer among heterogeneous machine learning models in multi-agent systems.

1.3 MOTIVATION

The increasing deployment of autonomous robots and intelligent agents in domains such as surveillance, smart transportation, industrial automation, disaster management, and autonomous navigation has created a growing demand for efficient collaboration among multiple agents [2], [6]. Although modern machine learning models have significantly improved perception capabilities, most multi-agent systems still operate with limited cooperation among heterogeneous models, resulting in redundant computation, increased latency, and inefficient utilization of available resources [3], [5]. Existing collaborative approaches such as federated learning and swarm learning primarily focus on model training and generally assume homogeneous model architectures, making them unsuitable for real-time knowledge sharing among heterogeneous agents [1], [4]. Similarly, knowledge distillation techniques require retraining and centralized processing, limiting their applicability in dynamic swarm environments [6], [10]. Consequently, lightweight agents are often unable to benefit

from the knowledge generated by more capable models, preventing the emergence of true collective intelligence within the swarm [7], [8].

This motivated the development of a swarm-intelligent and model-agnostic framework that enables heterogeneous machine learning models to collaborate through the exchange of high-level semantic knowledge rather than raw data or model parameters. By allowing agents to share structured knowledge packets and utilize collective memory, the proposed framework seeks to reduce redundant inference, improve resource efficiency, enhance perception performance, and promote decentralized collective intelligence in multi-agent systems [1], [2], [10]. The motivation behind this work is to bridge the gap between heterogeneous perception models and enable effective real-time collaboration in practical multi-robot environments.

1.4 OBJECTIVES OF THE PROJECT

1. To design and develop a decentralized swarm-intelligent framework that enables knowledge sharing among heterogeneous machine learning models in multi-robot systems.
2. To design a knowledge packet mechanism that captures object labels, confidence scores, timestamps, and agent information for inter-agent knowledge sharing.
3. To implement a collective memory framework that stores and distributes the highest-confidence perception knowledge among agents, reducing redundant inference and storage.
4. To evaluate the impact of knowledge sharing on model performance, knowledge reuse, and inference efficiency through comparative analysis before and after knowledge transfer.

1.5 SCOPE OF WORK

The scope of this work is focused on developing and evaluating a swarm-intelligent framework that enables knowledge transfer among heterogeneous machine learning models operating in a multi-agent environment. The framework is designed to support collaboration among agents employing different perception models, allowing them to share high-level semantic knowledge through structured knowledge packets instead of exchanging raw sensor data or model parameters [1], [6]. The implementation considers three heterogeneous deep learning models,

namely MobileNet, EfficientNet, and ResNet, acting as independent perception agents. The proposed framework facilitates knowledge sharing through a collective memory mechanism, where the most reliable knowledge packet is selected based on confidence scores and made available to other agents for reuse. This approach aims to reduce redundant inference, improve resource utilization, and enhance collaborative perception without requiring retraining or architectural modifications to the participating models [2], [10]. The work is limited to inference-level knowledge transfer using video-based human detection as a proof-of-concept application. The study primarily evaluates the effectiveness of knowledge packet generation, collective memory storage, knowledge reuse, and performance improvement among heterogeneous models. Although the framework is demonstrated in a simulated environment, the underlying architecture is sufficiently generic to be extended to real-world multi-robot systems, autonomous vehicles, distributed surveillance networks, smart environments, and other collaborative AI applications [6], [7], [11].

Future extensions may include support for larger swarms, multiple object classes, real-time robotic deployment, advanced knowledge fusion strategies, and adaptive communication mechanisms to further improve swarm-level intelligence and scalability [2], [10].

1.6 RESEARCH METHODOLOGY

The research methodology adopted in this work follows a design-oriented approach aimed at developing a decentralized and swarm-intelligent framework for knowledge transfer among heterogeneous machine learning models. The study began with an extensive review of existing literature on swarm intelligence, federated learning, distributed machine learning, multi-agent systems, and heterogeneous model collaboration to identify current limitations and research gaps [1], [2], [6].

Based on the identified gaps, a model-agnostic knowledge-sharing framework was designed in which multiple heterogeneous perception models operate as independent agents within a swarm environment. The proposed system employs MobileNet, EfficientNet, and ResNet as heterogeneous agents that independently process video frames and generate perception results. Instead of exchanging raw sensor data or model parameters, each agent converts its inference output into a structured knowledge packet containing semantic information such as object labels, confidence scores, timestamps, and agent identifiers [1], [10]. The generated knowledge

packets are stored in a Collective Memory module that serves as a shared knowledge repository. For each processed frame, the framework evaluates the generated packets and selects the packet with the highest confidence score as the most reliable representation of knowledge. This selected knowledge is then shared among participating agents, enabling weaker or less confident models to benefit from the knowledge generated by stronger agents without retraining or parameter exchange [2], [6].

To evaluate the effectiveness of the framework, experiments were conducted using video-based human detection scenarios. The performance of each model was analyzed before and after knowledge transfer using confidence-based evaluation metrics. The results were used to assess improvements in perception accuracy, knowledge reuse, reduction in redundant inference, and overall collaborative efficiency of the swarm [7], [10]. Through this methodology, the study demonstrates the feasibility of achieving collective intelligence among heterogeneous machine learning models through decentralized knowledge sharing and collective memory mechanisms.

1.7 ORGANIZATION OF THE REPORT

This dissertation is organized into six chapters to present the research work in a systematic and logical manner.

Chapter 1 – Introduction presents the background of the study, problem statement, motivation, objectives, scope of the work, research methodology, and overall organization of the report.

Chapter 2 – Literature Review discusses existing research related to swarm intelligence, multi-agent systems, federated learning, heterogeneous machine learning models, and knowledge-sharing mechanisms. It also presents a comparative analysis of existing approaches and identifies the research gap addressed by this work.

Chapter 3 – System Analysis and Design describes the existing system, proposed system, system architecture, design methodology, algorithms and models used, and the flowchart representing the overall system operation.

Chapter 4 – Implementation explains the development environment, hardware and software requirements, module descriptions, implementation details, and important aspects of the developed framework.

Chapter 5 – Results and Discussion presents the experimental setup, performance metrics, execution results, performance analysis, and discussion of the effectiveness of the proposed swarm-intelligent knowledge transfer framework.

Chapter 6 – Conclusion and Future Work summarizes the work carried out, highlights the major achievements and contributions, discusses the limitations of the current implementation, and suggests possible directions for future enhancements and research.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Swarm intelligence and multi-agent systems have emerged as important research areas for developing scalable, adaptive, and decentralized intelligent systems. Inspired by the collective behavior of biological swarms such as ants, bees, and bird flocks, swarm-intelligent systems enable multiple autonomous agents to cooperate and solve complex problems through local interactions without centralized control [2], [9]. Such systems have found applications in robotics, autonomous vehicles, wireless sensor networks, surveillance systems, Internet of Things (IoT) environments, and distributed artificial intelligence platforms [3], [5], [7].

The rapid advancement of machine learning and deep learning technologies has further enhanced the capabilities of autonomous agents by enabling sophisticated perception, classification, and decision-making functions [5], [11], [12]. However, practical multi-agent environments often consist of heterogeneous agents employing different machine learning models due to variations in hardware resources, computational capabilities, and application requirements [6], [8]. While model heterogeneity improves flexibility and deployment efficiency, it also introduces significant challenges in collaboration and knowledge sharing among agents. Several approaches such as federated learning, swarm learning, knowledge distillation, and distributed optimization have been proposed to facilitate collaboration among intelligent agents [1], [4], [10]. Although these approaches provide valuable mechanisms for distributed learning and coordination, most of them either assume homogeneous model architectures, require centralized training, or focus primarily on model optimization rather than real-time inference-level knowledge sharing. Consequently, redundant computation, inefficient resource utilization, and limited swarm-level intelligence continue to remain significant challenges in heterogeneous multi-agent systems [1], [6].

This chapter presents a comprehensive review of existing literature related to swarm intelligence, distributed learning, heterogeneous machine learning systems, knowledge-sharing mechanisms, and collaborative perception frameworks. The review analyzes the strengths and limitations of current approaches, compares their characteristics, and identifies

the research gap that motivates the proposed swarm-intelligent knowledge transfer framework for heterogeneous machine learning models.

2.2 REVIEW OF EXISTING LITERATURE

The development of intelligent multi-agent systems has been significantly influenced by advancements in swarm intelligence, distributed artificial intelligence, federated learning, collaborative machine learning, and decentralized optimization. Researchers have increasingly explored mechanisms that allow multiple autonomous agents to cooperate, exchange information, and collectively solve complex tasks while maintaining scalability and adaptability. Despite considerable progress, the challenge of enabling effective knowledge transfer among heterogeneous machine learning models remains largely unresolved. This section reviews the most relevant contributions in the areas of swarm intelligence, distributed learning, heterogeneous systems, and collaborative knowledge sharing.

Yang et al. [1] proposed an efficient UAV swarm-based multi-task federated learning framework designed for dynamic aerial environments. The framework enables multiple UAVs to collaboratively learn task-specific knowledge while preserving data privacy through decentralized training. A dynamic task allocation mechanism and task-affinity-based knowledge-sharing strategy were introduced to improve convergence speed and learning efficiency. Experimental results demonstrated enhanced task execution, reduced communication overhead, and improved utilization of swarm resources. The study highlights the effectiveness of integrating swarm intelligence with federated learning to enable cooperation among multiple autonomous agents. However, the framework assumes homogeneous learning architectures across participating UAVs and relies heavily on federated training procedures. Knowledge sharing occurs primarily during model training rather than during real-time inference. Consequently, the framework cannot efficiently support collaboration among heterogeneous machine learning models operating with different architectures. This work provides valuable insights into decentralized coordination, communication-efficient collaboration, and swarm-level learning. Our proposed framework adopts the concept of decentralized collaboration while extending it toward model-agnostic inference-level knowledge transfer among heterogeneous agents without requiring federated retraining.

Kazim et al. [2] presented a comprehensive review of Multi-Agent Swarm Engineering Systems (MASES), focusing on design principles, deployment strategies, scalability challenges, and coordination mechanisms. The study identified decentralization, self-organization, robustness, adaptability, and fault tolerance as the fundamental characteristics of successful swarm systems. Furthermore, the authors discussed the integration of artificial intelligence techniques to improve swarm decision-making and adaptability in large-scale autonomous environments. The work serves as an important reference for designing robust swarm architectures capable of operating under dynamic and uncertain conditions. However, its primary focus remains at the system architecture level and does not provide practical mechanisms for transferring learned knowledge between heterogeneous machine learning models. Our project draws inspiration from the decentralized architecture and coordination principles proposed in this study. We extend these concepts by introducing a semantic knowledge-sharing layer based on structured knowledge packets and collective memory, enabling heterogeneous agents to exchange perception knowledge efficiently.

Saadouni et al. [3] conducted a systematic review of intrusion detection systems that combine bio-inspired optimization algorithms with machine learning techniques in IoT environments. The study analyzed numerous swarm-based approaches and reported significant improvements in feature selection, classification accuracy, computational efficiency, and scalability. Techniques such as Particle Swarm Optimization, Ant Colony Optimization, and Genetic Algorithms were shown to enhance the performance of machine learning models in distributed environments. The review demonstrates how swarm intelligence can effectively improve distributed decision-making and optimization processes. Nevertheless, most surveyed approaches remain domain-specific and focus primarily on improving model performance rather than enabling collaborative knowledge sharing among heterogeneous agents. The findings reinforce the importance of swarm-inspired collaboration and distributed intelligence. Our framework extends these concepts beyond cybersecurity applications and utilizes swarm principles to facilitate generalized knowledge exchange among heterogeneous perception models.

Srinivasan et al. [4] proposed a federated meta-heuristic cryptography framework for secure cloud data migration. The framework integrates fuzzy logic, machine learning, reinforcement learning, and swarm-inspired optimization techniques to improve data security, reliability, and

communication efficiency. Experimental results demonstrated significant improvements in throughput, packet delivery ratio, and secure data transfer performance. Although the framework highlights the benefits of decentralized learning paradigms, its primary objective is secure cloud communication rather than collaborative perception or knowledge transfer. Additionally, it assumes relatively uniform learning structures and does not address heterogeneous machine learning environments. The decentralized philosophy presented in this work influenced the design of our framework. However, our approach focuses specifically on enabling cross-model knowledge exchange without requiring shared training processes, cryptographic synchronization, or parameter sharing.

Nizamudeen [5] introduced a swarm-based deep learning framework for intrusion detection in multi-cloud IoT environments. The proposed system combines swarm optimization techniques with deep neural networks to improve classification accuracy and adaptability in distributed environments. The framework achieved high detection performance while maintaining low false-positive rates. This study demonstrates the effectiveness of swarm intelligence in enhancing deep learning systems. However, the framework is designed for a specific security application and employs homogeneous learning models optimized for a single task. It does not investigate knowledge reuse or collaboration among heterogeneous machine learning architectures. The work contributes valuable insights regarding swarm-assisted intelligence enhancement. Our framework generalizes these concepts and applies them to task-independent, model-agnostic knowledge transfer among multiple heterogeneous agents.

He et al. [6] presented a survey of multimodal large-model-driven perception systems for UAV swarms operating in complex low-altitude environments. The study emphasizes adaptive edge intelligence, collaborative sensing, federated evolution, and swarm-level perception. The authors identified data heterogeneity, scalability, online adaptation, and collaborative intelligence as major research challenges in next-generation UAV systems. One of the most significant observations from this survey is the growing heterogeneity of perception models in autonomous systems. However, collaboration is primarily discussed in terms of coordinated sensing and perception rather than explicit semantic knowledge exchange. This work strongly motivates our research by validating the need for collaboration among heterogeneous learning models. Our framework addresses the identified gap through a structured knowledge packet mechanism that allows heterogeneous agents to exchange perception knowledge efficiently.

Zhang et al. [7] proposed an energy-efficient routing framework for heterogeneous wireless sensor networks using hybrid Particle Swarm Optimization and dynamic clustering. The framework dynamically adapts routing decisions according to network conditions, resulting in improved energy balancing and extended network lifetime. Although the study focuses on communication efficiency rather than machine learning, it demonstrates the value of adaptive collaboration among heterogeneous agents operating under resource constraints. Our project adopts similar resource-aware collaboration principles and applies them to knowledge-driven perception sharing among machine learning agents.

Abedpour et al. [8] developed a genetic-based clustering algorithm for resource allocation in heterogeneous fog computing platforms. The framework significantly reduced clustering errors, minimized latency, and improved overall resource utilization across distributed fog infrastructures. The study highlights effective techniques for managing heterogeneity through decentralized optimization. However, it does not address semantic knowledge transfer or collaborative learning among machine learning models. Our framework adopts the heterogeneity-aware design philosophy proposed in this work and extends it toward semantic knowledge exchange between heterogeneous perception agents.

Achouri et al. [9] combined swarm intelligence algorithms with neural networks for structural health monitoring applications. The proposed hybrid framework achieved superior prediction accuracy and convergence characteristics by integrating optimization techniques with deep learning models. The study illustrates how swarm intelligence can complement machine learning systems to improve robustness and performance. However, the approach remains centralized and application-specific, limiting its applicability to distributed collaborative environments. Our work generalizes this synergy and applies it to decentralized, real-time perception systems involving multiple autonomous agents.

Amico et al. [10] introduced a decentralized reinforcement learning framework that enables multiple language models to exchange learning experiences through shared rollouts. The proposed approach demonstrated substantial performance improvements without relying on centralized parameter synchronization. This work is particularly relevant because it introduces the concept of sharing high-level experience rather than raw data or model parameters. The findings demonstrate that semantic-level knowledge sharing can significantly improve learning efficiency. Our framework adopts a similar philosophy by enabling heterogeneous perception

models to exchange structured knowledge packets instead of raw sensor data or model parameters.

Sayed et al. [11] proposed an intelligent waste classification framework integrating deep learning with multi-objective Beluga Whale Optimization. Experimental evaluation demonstrated improvements in classification accuracy and computational efficiency. Although the framework validates the effectiveness of swarm-assisted intelligence enhancement, it remains limited to a specific application domain and employs centralized learning mechanisms. The study reinforces the value of swarm intelligence for improving learning systems, while our framework extends the concept toward decentralized collaborative knowledge sharing.

Kishore et al. [12] developed a Particle Swarm Optimization-based BiLSTM framework for intrusion detection in Internet of Vehicle environments. The integration of PSO significantly improved learning performance and attack detection accuracy. Despite its effectiveness, the framework does not support heterogeneous model collaboration or cross-agent knowledge reuse. Knowledge remains confined to individual learning models. Our work extends these swarm-enhanced learning concepts by introducing a model-agnostic mechanism capable of transferring knowledge among heterogeneous machine learning agents.

Rao et al. [13] proposed an intelligent security framework for Industrial IoT environments using swarm-optimized ensemble machine learning models. Experimental results demonstrated superior detection accuracy and improved system robustness compared with conventional approaches. While the framework effectively combines swarm intelligence and ensemble learning, its focus remains on application-specific optimization rather than knowledge transfer among heterogeneous agents. The study further validates the effectiveness of swarm-assisted intelligence and motivates the development of collaborative knowledge-sharing mechanisms capable of supporting diverse machine learning architectures.

The reviewed literature demonstrates substantial progress in swarm intelligence, federated learning, distributed artificial intelligence, optimization, and collaborative machine learning. Existing approaches successfully improve scalability, robustness, resource utilization, and learning performance. However, several critical limitations remain unresolved. Most existing frameworks assume homogeneous model architectures, rely on parameter synchronization, require collaborative retraining, or focus on optimization rather than knowledge reuse.

Furthermore, real-time inference-level collaboration among heterogeneous machine learning models is largely absent from current research. Therefore, a significant research gap exists in developing a decentralized, model-agnostic framework capable of enabling heterogeneous machine learning models to exchange semantic knowledge without sharing raw data, training datasets, or internal model parameters. Addressing this gap forms the primary motivation for the proposed swarm-intelligent framework based on structured knowledge packets, collective memory, and collaborative knowledge reuse among heterogeneous agents.

2.3 COMPARATIVE ANALYSIS OF EXISTING METHODS

Table 2.3 presents a comparative analysis of existing approaches related to swarm intelligence, federated learning, distributed optimization, heterogeneous systems, and collaborative machine learning.

Ref.	Approach	Strengths	Limitations
1.	UAV Swarm-based Federated Learning	Decentralized learning and dynamic task sharing	Assumes homogeneous models and focuses on training
2.	Swarm Engineering Systems	Scalable and decentralized swarm architecture	Does not address ML knowledge sharing
3.	Bio-inspired IoT Intrusion Detection	Distributed intelligence and optimization	Domain-specific and lacks heterogeneous collaboration
4.	Federated Meta-Heuristic Framework	Privacy-preserving decentralized learning	Focuses on optimization rather than perception
5.	Swarm-based Deep Learning IDS	Improved classification performance	Uses homogeneous models and application-specific design
6.	Multimodal UAV Perception Survey	Highlights heterogeneous perception environments	Does not provide structured knowledge-sharing mechanisms
7.	PSO-based Heterogeneous Routing	Resource-efficient collaboration	Focuses on networking rather than machine learning

8.	Genetic-based Fog Clustering	Supports heterogeneity and resource management	Does not address perception knowledge transfer
9.	Swarm Intelligence for Structural Monitoring	Improved robustness and optimization	Centralized and application-specific
10.	Collective Experience Sharing for LMs	Shares high-level knowledge efficiently	Limited to language models and training scenarios
11.	Swarm-Optimized Waste Classification	Enhanced learning performance	Single-domain application and centralized processing
12.	PSO-BiLSTM for IoV Security	Improved distributed learning efficiency	No heterogeneous model collaboration
13.	Swarm-Optimized Ensemble ML Framework	Intelligent distributed decision-making	Does not support model-agnostic knowledge transfer

The comparative study reveals that existing approaches successfully utilize swarm intelligence, federated learning, optimization techniques, and distributed machine learning to improve collaboration among intelligent agents. However, most of the reviewed methods assume homogeneous learning models, focus on collaborative training, or are designed for specific application domains [1], [4], [5]. Very few studies address real-time inference-level knowledge sharing among heterogeneous machine learning models [6], [10]. Furthermore, existing frameworks generally rely on model-specific communication mechanisms and do not provide a universal representation for exchanging perception knowledge across different architectures [2], [7], [8]. These limitations motivate the development of the proposed swarm-intelligent framework, which enables model-agnostic knowledge transfer through structured knowledge packets and collective memory mechanisms.

2.4 RESEARCH GAP IDENTIFICATION

The literature survey indicates that significant progress has been made in the areas of swarm intelligence, federated learning, distributed optimization, and collaborative machine learning systems. Existing studies have demonstrated the benefits of decentralized decision-making, swarm-based coordination, and distributed learning for improving scalability, adaptability, and system performance in multi-agent environments [1], [2], [5], [9]. However, several important research gaps remain unresolved.

Most existing collaborative learning approaches, including federated learning and swarm learning, primarily focus on model training and generally assume homogeneous machine learning architectures across participating agents [1], [4]. Such assumptions limit their applicability in real-world multi-robot systems where agents often employ different machine learning models based on varying computational capabilities and hardware constraints [6], [8]. Another major limitation is that existing approaches concentrate on parameter sharing, gradient aggregation, or centralized knowledge distillation processes, which require retraining and are not suitable for real-time inference-level collaboration [1], [4], [10]. Consequently, agents continue to perform redundant inference even when similar knowledge has already been generated elsewhere in the swarm, leading to increased computational overhead, latency, and resource consumption [3], [5].

Although several studies discuss heterogeneity in distributed systems, sensor networks, and fog computing environments, they primarily focus on resource allocation, routing, clustering, or optimization rather than semantic knowledge exchange among heterogeneous machine learning models [7], [8]. Similarly, recent works on swarm engineering and collaborative perception emphasize communication and coordination but do not provide a generalized mechanism for sharing perception knowledge across different model architectures [2], [6].

Furthermore, existing solutions lack a standardized and model-agnostic representation that enables heterogeneous agents to interpret, store, and reuse knowledge generated by other agents. The absence of such a mechanism prevents the emergence of true collective intelligence, where agents can continuously benefit from the experiences and observations of the entire swarm [6], [10], [13].

Therefore, there exists a clear research gap in developing a decentralized framework that supports real-time, inference-level, model-agnostic knowledge transfer among heterogeneous

machine learning models. Addressing this gap forms the primary motivation for the proposed work, which introduces structured knowledge packets and a collective memory mechanism to enable efficient knowledge sharing, reduce redundant inference, improve resource utilization, and promote swarm-level collective intelligence in multi-agent systems.

2.5 SUMMARY

This chapter presented a comprehensive review of existing research related to swarm intelligence, multi-agent systems, federated learning, distributed machine learning, and heterogeneous collaborative environments. The reviewed studies demonstrated the effectiveness of decentralized coordination, swarm-based optimization, distributed intelligence, and collaborative learning in improving system performance across various application domains such as robotics, IoT, cloud computing, wireless sensor networks, and autonomous systems [1], [2], [5], [7].

The literature analysis revealed that most existing approaches focus primarily on collaborative training, optimization, resource management, or domain-specific applications. While these methods successfully support decentralized operation and distributed decision-making, they generally assume homogeneous machine learning models and do not adequately address real-time knowledge sharing among heterogeneous perception systems [1], [4], [6]. Furthermore, existing frameworks often rely on parameter sharing, retraining, or centralized processing mechanisms, which limit their applicability in dynamic multi-robot environments [3], [10].

The comparative analysis and research gap identification highlighted the absence of a model-agnostic framework capable of enabling inference-level knowledge transfer among heterogeneous machine learning models. This gap motivated the development of the proposed swarm-intelligent knowledge transfer framework, which utilizes structured knowledge packets and collective memory to facilitate decentralized collaboration without sharing raw sensor data or model parameters.

The findings of this chapter provide the theoretical foundation for the proposed work and justify the need for a scalable, efficient, and model-independent knowledge-sharing mechanism. Based on these observations, the next chapter presents the system analysis and design of the proposed framework, including the architecture, methodology, models used, and operational workflow.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 EXISTING SYSTEM

Existing multi-agent and swarm-intelligent systems primarily rely on collaborative learning approaches such as Federated Learning, Swarm Learning, Knowledge Distillation, and Distributed Optimization techniques to enable cooperation among multiple intelligent agents [1], [4], [10]. These approaches have demonstrated significant improvements in distributed learning, privacy preservation, and decentralized decision-making across domains such as robotics, IoT, wireless sensor networks, and autonomous systems [2], [5], [7].

Federated Learning enables multiple agents to collaboratively train a shared machine learning model without exchanging raw data. Instead, model parameters or gradients are periodically aggregated to improve overall learning performance [1], [4]. Similarly, Swarm Learning extends this concept by incorporating decentralized coordination mechanisms inspired by swarm intelligence principles. These approaches successfully preserve privacy and reduce centralized dependency; however, they generally assume homogeneous model architectures across participating agents, limiting their applicability in heterogeneous environments [1].

Knowledge Distillation is another widely adopted approach in which a larger model transfers its learned knowledge to a smaller model through a training process [10]. Although effective for model compression and performance improvement, knowledge distillation requires retraining and offline processing, making it unsuitable for dynamic real-time collaboration among autonomous agents. Furthermore, the transferred knowledge is typically unidirectional and does not support continuous knowledge exchange among multiple heterogeneous agents. Several swarm-based systems have also been proposed for intrusion detection, resource allocation, routing optimization, cloud computing, and autonomous decision-making [3], [5], [7], [8], [12], [13]. While these systems demonstrate the effectiveness of decentralized intelligence and swarm coordination, they are generally application-specific and focus on optimization or training rather than real-time perception knowledge sharing. As a result, agents continue to perform independent inference even when similar knowledge is already available within the system.

The major limitations of existing systems can be summarized as follows:

Dependence on homogeneous machine learning architectures. Focus on collaborative training rather than inference-level collaboration. Requirement of parameter sharing, gradient exchange, or retraining. Lack of a model-agnostic knowledge representation mechanism. Redundant inference performed by multiple agents on similar inputs. Inefficient utilization of collective intelligence in heterogeneous environments. Limited support for real-time knowledge reuse and decentralized perception sharing.

These limitations highlight the need for a decentralized and model-agnostic framework that enables heterogeneous machine learning models to exchange high-level semantic knowledge during inference. The proposed system addresses these challenges through structured knowledge packets and collective memory mechanisms, enabling efficient knowledge transfer and swarm-level collective intelligence.

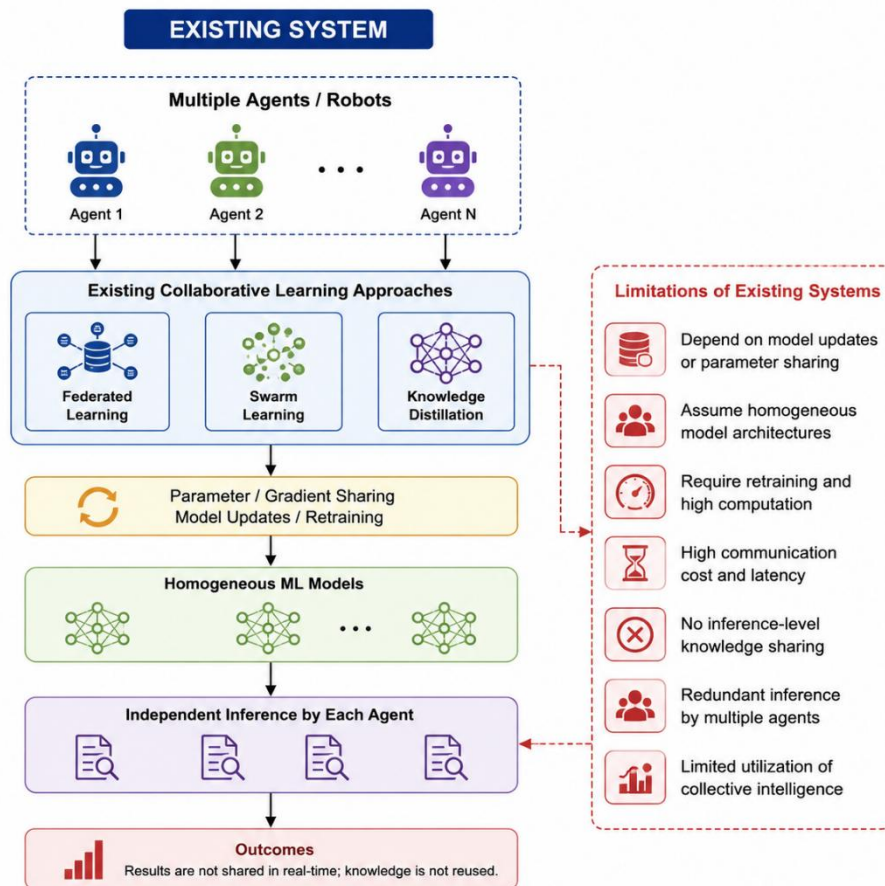


Figure 3.1: Existing System Architecture – Traditional Collaborative Learning Approaches and Their Limitations

Figure 3.1 illustrates the workflow followed by existing collaborative learning approaches such as Federated Learning, Swarm Learning, and Knowledge Distillation. In these systems, agents collaborate primarily through model updates, parameter sharing, or retraining mechanisms. Most approaches assume homogeneous machine learning models and focus on training-level collaboration rather than inference-level knowledge sharing. Consequently, agents continue to perform redundant inference independently, resulting in increased computational overhead, latency, and inefficient utilization of swarm intelligence. These limitations motivate the need for a decentralized and model-agnostic knowledge-sharing framework capable of supporting heterogeneous machine learning models in real-time environments.

3.2 PROPOSED SYSTEM

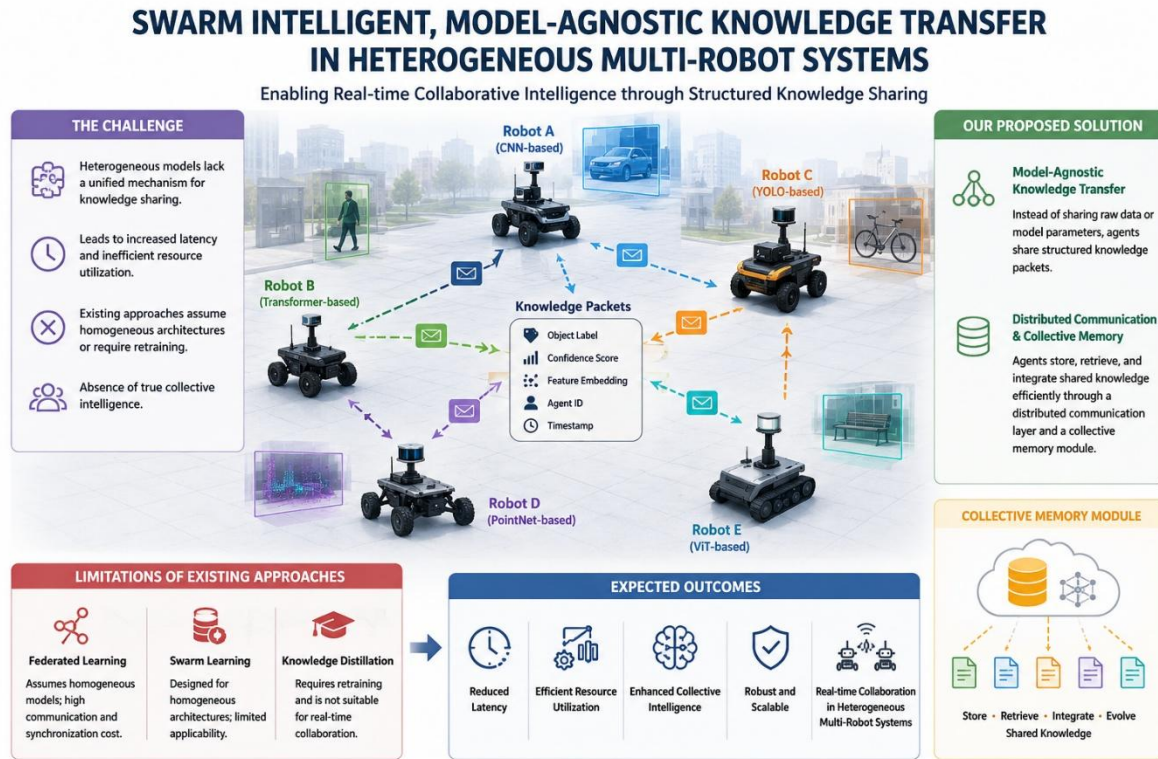


Figure 3.2: Proposed system for the swarm intelligent framework

The proposed system introduces a decentralized swarm-intelligent framework that enables knowledge transfer among heterogeneous machine learning models operating as autonomous agents. Unlike traditional systems that rely on centralized processing, homogeneous models, or parameter sharing, the proposed framework allows different machine learning models to

collaborate by exchanging high-level semantic knowledge during inference. The system employs three heterogeneous perception agents based on MobileNet, EfficientNet, and ResNet architectures, each independently processing video frames and generating prediction results. The framework begins with video acquisition and frame extraction. Each extracted frame is simultaneously processed by all three agents, which perform local inference to identify the presence of a human and generate corresponding confidence scores. The inference outputs are transformed into standardized knowledge packets containing the frame ID, object label, confidence score, timestamp, and source agent information. These packets provide a common representation that can be understood by all agents irrespective of their underlying model architecture.

The generated knowledge packets are stored in a Collective Memory module that serves as a shared knowledge repository for the swarm. For each frame, the packet with the highest confidence score is selected as the best knowledge representation and retained in memory. This best knowledge is then shared with other agents through a decentralized communication mechanism. The receiving agents utilize the shared knowledge to update their internal confidence and improve prediction quality without requiring retraining, parameter exchange, or raw data sharing.

By enabling knowledge reuse among heterogeneous models, the proposed system reduces redundant inference, improves resource utilization, enhances prediction accuracy, and promotes collaborative intelligence. The decentralized nature of the framework also improves scalability, fault tolerance, and adaptability, making it suitable for real-world multi-robot and distributed AI environments.

3.3 SYSTEM ARCHITECTURE

The proposed methodology develops a swarm-intelligent, model-agnostic knowledge transfer framework using three heterogeneous models: MobileNet, EfficientNet, and ResNet. Each model independently processes video frames and generates knowledge packets containing object labels, confidence scores, timestamps, and agent identifiers. These packets are stored in a Collective Memory module, where the highest-confidence packet for each frame is selected as the best knowledge. The selected knowledge is shared with other agents, enabling knowledge transfer without exchanging raw data or model parameters. Finally, model

performance is evaluated before and after knowledge transfer to assess improvements in collaborative perception and accuracy.

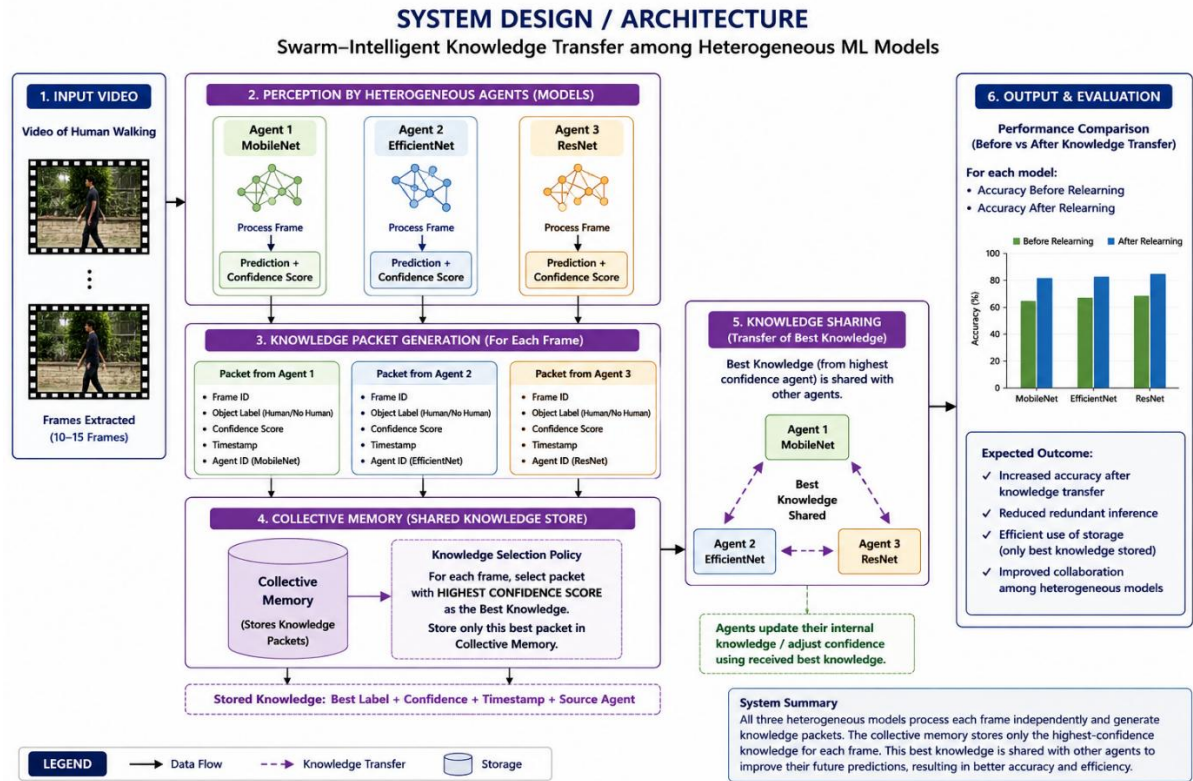


Figure 3.3: illustrates the overall architecture of the proposed Swarm-Intelligent Knowledge Transfer Framework for heterogeneous machine learning models.

The architecture consists of six major stages:

Input Video Acquisition

The system receives a video containing human walking activity as input. The video is divided into multiple frames, typically ranging from 10 to 15 frames, which are used for perception and knowledge-sharing operations. The extracted frames serve as common input for all heterogeneous agents participating in the swarm.

Perception by Heterogeneous Agents

The framework employs three heterogeneous machine learning models:

- Agent 1 – MobileNet
- Agent 2 – EfficientNet
- Agent 3 – ResNet

Each agent independently processes every video frame and performs human detection. Based on its learned representation and architecture, each model produces:

- Predicted object label
- Confidence score

Since the three models possess different feature extraction capabilities, their confidence scores may vary for the same frame. This stage reflects the heterogeneous nature of the swarm where multiple agents perceive the same environment using different machine learning architectures.

Knowledge Packet Generation

After completing inference on a frame, each agent generates a structured Knowledge Packet.

Each packet contains:

- Frame ID
- Object Label
- Confidence Score
- Timestamp
- Agent Identifier

The knowledge packet represents a semantic summary of the perception result and acts as the primary unit of knowledge exchange within the swarm. By sharing compact semantic information instead of raw image data, communication overhead is significantly reduced.

Collective Memory Module

The generated knowledge packets are transmitted to the Collective Memory module, which acts as a shared knowledge repository for all agents. For every processed frame, the collective memory evaluates all received knowledge packets and applies a knowledge-selection policy.

The packet with the highest confidence score is selected as the Best Knowledge Packet.

Only the best packet is stored in collective memory, while lower-confidence packets are discarded.

The stored information includes:

- Best Object Label
- Highest Confidence Score
- Timestamp
- Source Agent

This mechanism prevents redundant storage and ensures that only the most reliable perception knowledge is retained.

Knowledge Sharing and Transfer

Once the best knowledge packet is identified, it is shared with all participating agents.

Through this process:

- Stronger agents contribute reliable knowledge.
- Weaker agents receive improved perception information.
- Knowledge becomes available across heterogeneous models.

The receiving agents update their internal representations using the shared knowledge packet. This enables knowledge transfer among heterogeneous machine learning models without exchanging model parameters or training data. The process mimics swarm-intelligence behavior where local interactions collectively improve the performance of the entire system.

Output and Performance Evaluation

The final stage evaluates the effectiveness of the proposed framework.

For each model, performance is measured:

- Before Knowledge Transfer
- After Knowledge Transfer

The comparison demonstrates the impact of collaborative knowledge sharing on model performance.

The expected outcomes include:

- Improved accuracy after knowledge transfer
- Reduced redundant inference
- Efficient memory utilization
- Better collaboration among heterogeneous models
- Emergence of swarm-level collective intelligence

The overall operation of the system follows a decentralized swarm-intelligence approach. All heterogeneous agents independently analyze each frame and generate perception knowledge. The collective memory stores only the most reliable knowledge for each frame based on confidence scores. This best knowledge is subsequently shared with other agents, enabling them to improve their predictions and benefit from the strengths of other models.

As knowledge accumulates and is continuously reused, the swarm exhibits collective intelligence that exceeds the capability of any individual model. The proposed framework therefore demonstrates an effective mechanism for real-time, model-agnostic knowledge

transfer among heterogeneous machine learning models while maintaining decentralization, scalability, and communication efficiency.

The Knowledge Packet Generation stage enables knowledge sharing among heterogeneous machine learning models. After processing each video frame, every agent (MobileNet, EfficientNet, and ResNet) generates a knowledge packet containing the essential perception information obtained from its inference result.

Each knowledge packet consists of:

- Frame ID
- Object Label (Human/No Human)
- Confidence Score
- Timestamp
- Agent Identifier

The generated knowledge packets provide a compact semantic representation of the perception result and act as the primary unit of communication among agents. Instead of sharing raw images or model parameters, agents exchange these lightweight knowledge packets, thereby reducing communication overhead and preserving model independence. For every frame, the knowledge packets generated by all agents are forwarded to the Collective Memory module, where the packet with the highest confidence score is selected as the best knowledge representation. This selected packet is subsequently shared with other agents to facilitate knowledge transfer and collaborative perception.

The Collective Memory module serves as a shared knowledge repository for the proposed framework. After processing each video frame, MobileNet, EfficientNet, and ResNet agents generate individual knowledge packets containing the object label, confidence score, timestamp, and agent identifier. For every frame, the collective memory receives knowledge packets from all three agents and applies a knowledge selection policy. The packet with the highest confidence score is selected as the best knowledge representation for that frame. Only this selected packet is stored in the collective memory, while lower-confidence packets are discarded.

The stored knowledge includes:

- Object Label
- Confidence Score

- Timestamp
- Source Agent ID

The selected knowledge is then shared with the remaining agents, enabling them to utilize the most reliable perception result available within the swarm. This approach reduces redundant storage, improves knowledge reuse, and supports collaborative perception among heterogeneous machine learning models. By maintaining only the highest-confidence knowledge for each frame, the collective memory acts as an efficient mechanism for knowledge selection, storage, and sharing, contributing to improved overall system performance.

3.4 DESIGN METHODOLOGY

The design methodology of the proposed system follows a decentralized swarm-intelligence approach that enables knowledge transfer among heterogeneous machine learning models. The methodology is designed to facilitate collaboration during inference rather than training, allowing autonomous agents to share useful perception knowledge without exchanging raw data or model parameters.

The process begins with **video acquisition and frame extraction**, where the input video is divided into multiple frames for analysis. Each extracted frame is then processed independently by three heterogeneous agents based on MobileNet, EfficientNet, and ResNet architectures. These agents perform local inference and generate prediction outputs consisting of object labels and confidence scores.

The inference results are converted into **knowledge packets**, which contain essential information such as frame ID, object label, confidence score, timestamp, and source agent ID. These packets provide a standardized representation that can be interpreted by all agents irrespective of their underlying model architecture.

The generated packets are forwarded to the **Collective Memory** module, where a knowledge selection policy compares confidence scores from all agents. The packet with the highest confidence value is selected as the best knowledge representation for that frame and stored in the shared memory repository.

Once stored, the selected knowledge packet is disseminated to other agents through a decentralized communication mechanism. The receiving agents utilize the shared knowledge

to update or reinforce their predictions, thereby benefiting from the experience of other agents without requiring retraining or parameter exchange.

This process is repeated for all extracted frames. After processing is completed, the performance of each model is evaluated before and after knowledge transfer to measure improvements in accuracy and collaborative efficiency. Through continuous knowledge sharing and reuse, the framework enables the emergence of collective intelligence while reducing redundant computation and communication overhead.

3.5 ALGORITHMS/MODELS USED

The proposed swarm-intelligent framework employs three heterogeneous deep learning models as autonomous perception agents. These models were selected due to their diverse architectural characteristics, computational requirements, and performance capabilities, making them suitable for demonstrating knowledge transfer among heterogeneous machine learning systems.

MobileNet-V3 Small is a lightweight convolutional neural network designed for resource-constrained devices such as mobile and embedded systems. The model utilizes depthwise separable convolutions, squeeze-and-excitation modules, and neural architecture search techniques to achieve high efficiency with low computational cost. In the proposed framework, MobileNet acts as a lightweight perception agent capable of performing real-time inference while consuming minimal computational resources. Its role is particularly important in demonstrating how low-capacity agents can benefit from knowledge shared by more powerful models.

EfficientNet-B0 is a deep learning architecture that employs compound scaling to balance network depth, width, and input resolution. This approach enables EfficientNet to achieve superior accuracy while maintaining computational efficiency. Within the proposed framework, EfficientNet serves as a medium-complexity perception agent that provides a balance between inference speed and prediction accuracy. Its outputs contribute to the collective memory and knowledge-sharing process, enhancing overall swarm intelligence.

ResNet-18 is a residual neural network that introduces skip connections to overcome the vanishing gradient problem encountered in deep networks. These residual connections enable effective feature learning and improved classification performance. In the proposed system,

ResNet-18 functions as a high-capability perception agent capable of generating reliable predictions with strong feature representation. Its knowledge often contributes significantly to the collective memory due to its robust inference performance.

After local inference, each agent generates a standardized knowledge packet containing:

- Frame ID
- Object Label
- Confidence Score
- Timestamp
- Agent Identifier

This packet serves as the fundamental unit of information exchange within the swarm. The standardized format ensures interoperability among heterogeneous models and enables model-agnostic knowledge sharing.

The Collective Memory module employs a confidence-based selection strategy to identify the most reliable knowledge for each frame. The confidence scores generated by all agents are compared, and the packet with the highest confidence value is selected as the best knowledge representation.

The selection process can be expressed as:

$$BestPacket = \max (Confidence_1, Confidence_2, Confidence_3)$$

The selected packet is stored in collective memory and shared with other agents for knowledge reuse.

The framework utilizes a decentralized swarm-based knowledge-sharing mechanism in which the selected best packet is disseminated to other agents. Instead of exchanging raw images, feature maps, or model parameters, only semantic knowledge is shared. This approach minimizes communication overhead while enabling collaborative intelligence among heterogeneous models.

3.6 FLOWCHART

The system flowchart illustrates the complete workflow of the proposed swarm-intelligent knowledge transfer framework. The framework employs three heterogeneous machine

learning agents—MobileNet, EfficientNet, and ResNet—to collaboratively process video frames and share knowledge through a collective memory mechanism.

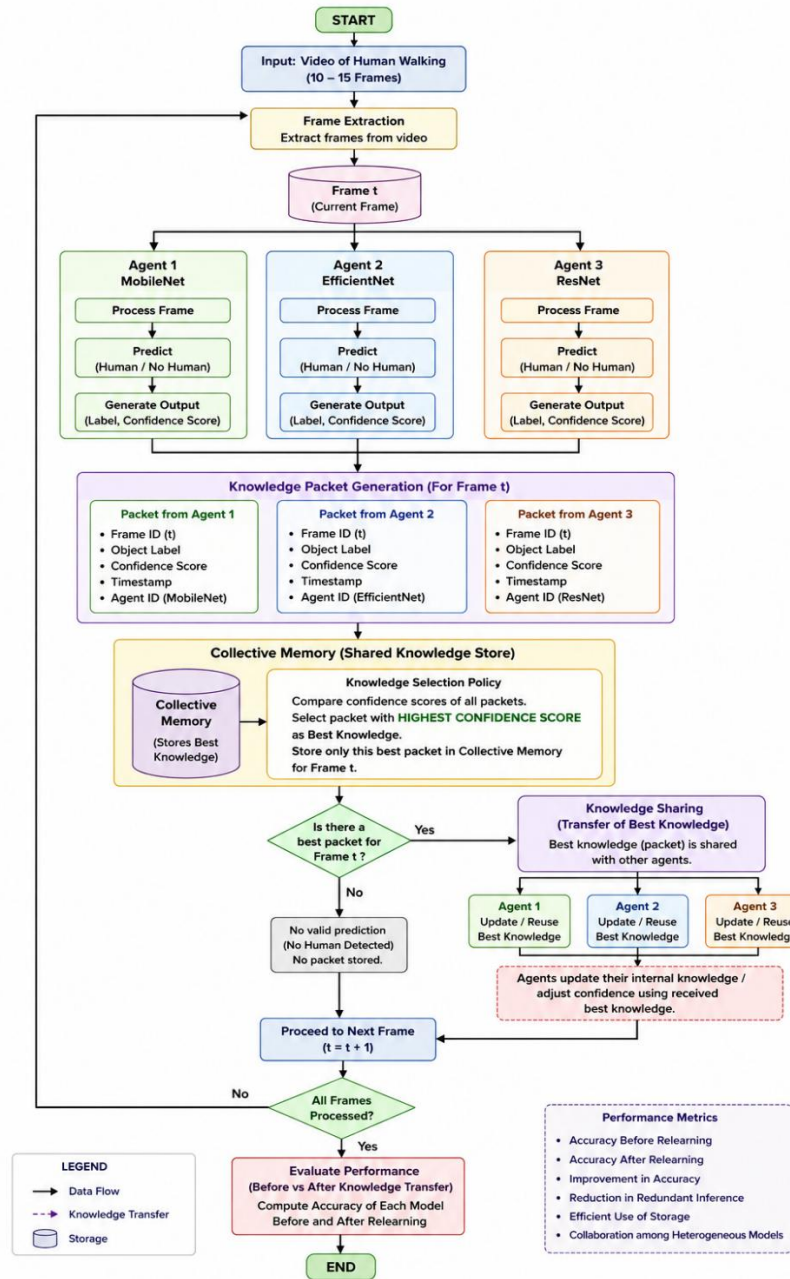


Figure 3.6: System Flowchart of the Proposed Swarm-Intelligent Knowledge Transfer Framework. The process begins with a video containing a human walking. The input video is divided into multiple frames (approximately 10–15 frames), which are processed sequentially. Each extracted frame is considered as the current frame for analysis.

For every frame, all three agents independently perform perception using their respective machine learning models:

- Agent 1: MobileNet
- Agent 2: EfficientNet
- Agent 3: ResNet

Each agent processes the frame and predicts whether a human is present. The prediction result consists of:

- Object Label (Human / No Human)
- Confidence Score

Since the agents use different architectures, their confidence scores may vary even when analyzing the same frame.

After inference, each agent generates a knowledge packet containing:

- Frame ID
- Object Label
- Confidence Score
- Timestamp
- Agent Identifier

These knowledge packets represent the perception knowledge generated by individual agents and serve as the basis for knowledge sharing.

All generated knowledge packets are forwarded to the Collective Memory module. The collective memory acts as a shared knowledge repository and applies a knowledge selection policy. For each frame, the confidence scores of all packets are compared. The packet with the highest confidence score is selected as the **Best Knowledge Packet** and stored in the collective memory, while lower-confidence packets are ignored. This ensures that only the most reliable knowledge is retained for each frame.

Once the best packet is identified, it is shared with the remaining agents. Through this process, agents gain access to the most confident prediction available within the swarm. The receiving agents update or reuse this shared knowledge, thereby improving their perception quality and demonstrating knowledge transfer among heterogeneous machine learning models. If no valid prediction is available for a frame, no packet is stored and the system proceeds directly to the next frame.

The above process is repeated for all extracted frames. For every frame:

1. All agents perform inference.
2. Knowledge packets are generated.
3. Collective memory selects the best packet.
4. Best knowledge is shared among agents.
5. The system moves to the next frame.

This iterative mechanism enables continuous collaboration and knowledge propagation throughout the swarm.

After all frames have been processed, the system evaluates the performance of each model before and after knowledge transfer. The evaluation focuses on:

- Accuracy before relearning
- Accuracy after relearning
- Accuracy improvement
- Reduction in redundant inference
- Efficient knowledge storage
- Collaboration among heterogeneous models

The flowchart demonstrates how decentralized knowledge sharing enables heterogeneous machine learning models to collaboratively improve perception performance without exchanging raw video data or model parameters, thereby achieving swarm-level intelligence.

CHAPTER 4

IMPLEMENTATION

4.1 DEVELOPMENT ENVIRONMENT

The proposed swarm-intelligent knowledge transfer framework was developed and tested in a Python-based environment for implementing heterogeneous machine learning models, collective memory management, and knowledge-sharing mechanisms. The development environment provides the necessary software libraries, deep learning frameworks, and execution tools required for model inference, knowledge packet generation, and performance evaluation.

The implementation was carried out using **Python 3.x**, which served as the primary programming language due to its extensive support for machine learning, computer vision, and distributed system development. Deep learning models such as MobileNet-V3 Small, EfficientNet-B0, and ResNet-18 were implemented using the **PyTorch** framework, which provides pre-trained models, efficient tensor operations, and GPU acceleration capabilities. Video processing and frame extraction were performed using the **OpenCV** library, while data handling and numerical computations were supported by **NumPy** and **Pandas**.

The framework utilizes a **Collective Memory** module to store and retrieve knowledge packets generated by different agents. Inter-agent communication and memory synchronization were implemented using lightweight distributed data structures and shared memory mechanisms. Experimental evaluation and visualization of results were carried out using **Matplotlib** and **Seaborn** libraries. The development and testing of the system were performed using **Visual Studio Code (VS Code)** as the integrated development environment (IDE). The complete implementation was executed on a Windows-based system with sufficient computational resources to support deep learning inference and video processing tasks.

Table 4.1: Development environment

Component	Specification
Programming Language	Python 3.x
IDE	Visual Studio Code (VS Code)
Deep Learning Framework	PyTorch
Computer Vision Library	OpenCV

Component	Specification
Data Processing Libraries	NumPy, Pandas
Visualization Tools	Matplotlib, Seaborn
Operating System	Windows 10/11
Version Control	Git (Optional)

4.2 HARDWARE REQUIREMENTS

The proposed framework was implemented and evaluated using a standard computing system capable of supporting deep learning inference, video processing, and knowledge-sharing operations. Since the framework employs pre-trained deep learning models and performs frame-by-frame analysis, moderate computational resources are sufficient for execution. The hardware configuration used during experimentation is summarized in Table 4.2.

Table 4.2: Hardware requirements

Hardware Component	Specification
Processor	Intel Core i5/i7 or equivalent
RAM	Minimum 8 GB (16 GB Recommended)
Storage	Minimum 20 GB Free Disk Space
Display	1366 × 768 Resolution or Higher
Input Device	Keyboard and Mouse

The proposed system does not require specialized hardware and can be executed on standard desktop or laptop computers. The use of a dedicated GPU can accelerate inference and video processing; however, the framework is designed to operate efficiently even on CPU-based systems. This makes the solution suitable for resource-constrained environments and heterogeneous computing platforms.

4.3 SOFTWARE REQUIREMENTS

The implementation of the proposed swarm-intelligent knowledge transfer framework requires a software environment capable of supporting deep learning inference, computer vision operations, data processing, and performance evaluation. Various software tools, frameworks, and libraries were utilized to develop, execute, and analyze the system. Python was used as the primary programming language due to its simplicity, extensive library support, and suitability

for machine learning applications. The deep learning models, namely MobileNet-V3 Small, EfficientNet-B0, and ResNet-18, were implemented using the PyTorch framework. OpenCV was employed for video processing, frame extraction, and image manipulation. Additional libraries such as NumPy and Pandas were used for numerical computation and data handling, while Matplotlib and Seaborn were utilized for result visualization and performance analysis.

Table 4.3: Software requirements

Software Component	Purpose
Windows 10/11	Operating System
Python 3.x	Programming Language
Visual Studio Code (VS Code)	Development Environment (IDE)
PyTorch	Deep Learning Framework
OpenCV	Video and Image Processing
NumPy	Numerical Computation
Pandas	Data Processing and Analysis
Matplotlib	Graph and Result Visualization
Seaborn	Performance Visualization
Fakeredis	Collective Memory Simulation
JSON	Knowledge Packet Storage and Exchange

The software environment was selected to ensure efficient implementation of heterogeneous machine learning models, knowledge packet generation, collective memory management, and performance evaluation. The use of open-source tools and libraries enhances portability, scalability, and ease of future system extensions.

4.4 MODULE DESCRIPTION

The proposed swarm-intelligent knowledge transfer framework is implemented using a modular architecture, where each module performs a specific function in the overall knowledge-sharing process. The modules work together to enable collaboration among heterogeneous machine learning models while maintaining decentralized operation and model independence.

Video Acquisition and Frame Extraction Module

This module is responsible for accepting the input video and extracting individual frames for analysis. The extracted frames serve as the input data for all perception agents. By processing frames individually, the system can evaluate model performance and knowledge transfer on a frame-by-frame basis.

Functions:

- Load input video.
- Extract video frames.
- Preprocess frames for model inference.
- Supply frames to all perception agents.

Heterogeneous Perception Agent Module

This module consists of three independent perception agents based on MobileNet-V3 Small, EfficientNet-B0, and ResNet-18. Each agent processes the same frame independently and generates prediction results along with confidence scores.

Functions:

- Perform local inference.
- Detect the target object (human).
- Generate confidence scores.
- Operate independently without agent dependency.

Knowledge Packet Generation Module

After inference, each agent converts its prediction result into a standardized knowledge packet. The packet contains only high-level semantic information required for knowledge sharing.

Knowledge Packet Fields:

- Frame ID
- Object Label
- Confidence Score
- Timestamp
- Agent ID

Functions:

- Extract semantic information.
- Create structured knowledge packets.

- Ensure model-independent representation.

Collective Memory Module

The Collective Memory module acts as a shared knowledge repository where packets generated by all agents are stored and managed. For each frame, the packet with the highest confidence score is selected and retained as the best knowledge representation.

Functions:

- Store knowledge packets.
- Compare confidence scores.
- Select the best packet.
- Maintain shared swarm knowledge.

Knowledge Sharing Module

This module enables the selected knowledge packet to be shared among all participating agents. Instead of sharing images or model weights, only semantic knowledge is exchanged.

Functions:

- Distribute best knowledge packet.
- Enable inter-agent collaboration.
- Reduce communication overhead.
- Support decentralized knowledge transfer.

Knowledge Integration Module

The receiving agents utilize the shared knowledge packet and integrate it with their local predictions. This process helps weaker models benefit from knowledge generated by stronger models.

Functions:

- Receive shared knowledge.
- Update prediction confidence.
- Improve decision quality.
- Support collective intelligence.

Performance Evaluation Module

This module evaluates the effectiveness of the proposed framework by comparing model performance before and after knowledge transfer.

Functions:

- Calculate model accuracy.
- Measure improvement after knowledge sharing.
- Generate comparison graphs.
- Analyze swarm intelligence performance.

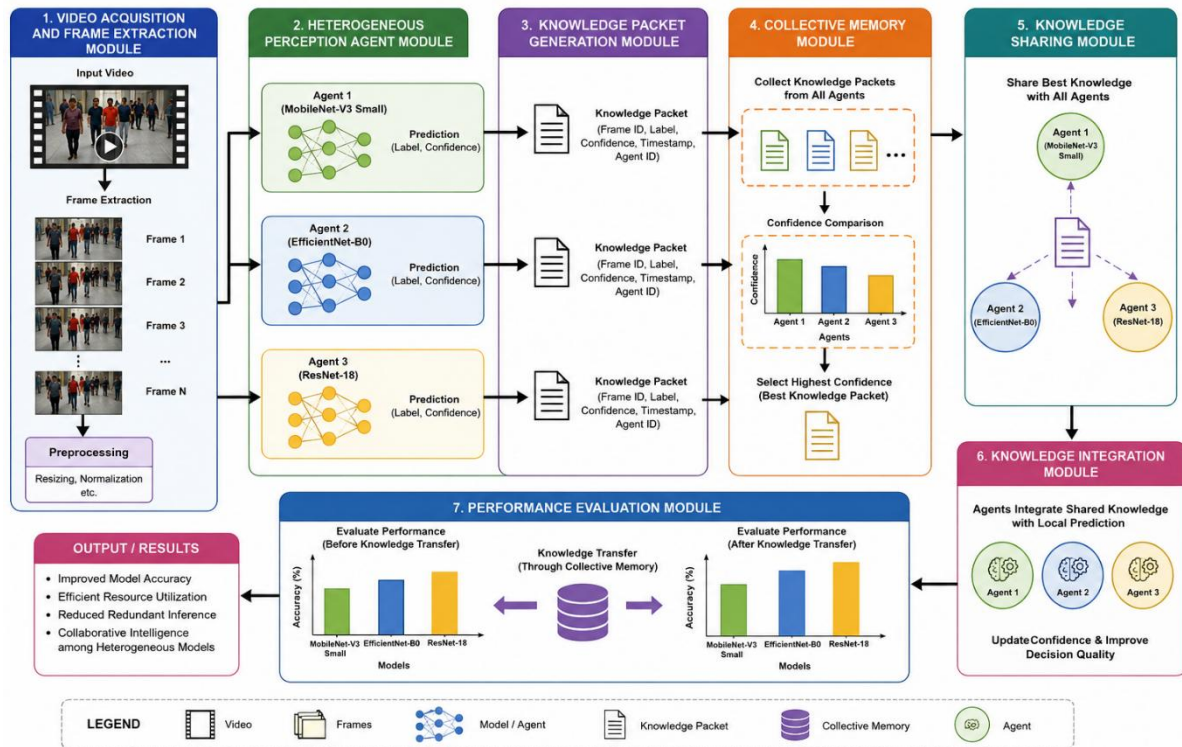


Figure 4.4: Functional Modules of the Proposed Swarm-Intelligent Knowledge Transfer Framework

Figure 4.4: Functional Modules of the Proposed Swarm-Intelligent Knowledge Transfer Framework

Figure 4.4 illustrates the interaction among the major modules of the system, including frame extraction, heterogeneous perception agents, knowledge packet generation, collective memory, knowledge sharing, knowledge integration, and performance evaluation. Together, these modules enable decentralized knowledge transfer and collaborative intelligence among heterogeneous machine learning models.

4.5 IMPLEMENTATION DETAILS

The proposed swarm-intelligent knowledge transfer framework was implemented using Python and a collection of open-source machine learning and computer vision libraries. The implementation focuses on enabling collaboration among heterogeneous machine learning

models through knowledge packet generation, collective memory storage, and decentralized knowledge sharing.

The implementation begins with the **video processing module**, where an input video is provided to the system. Using OpenCV, the video is decomposed into individual frames that serve as input for the perception agents. Each extracted frame is preprocessed through resizing and normalization to ensure compatibility with the deep learning models.

Three heterogeneous perception agents based on **MobileNet-V3 Small, EfficientNet-B0, and ResNet-18** are initialized using pre-trained PyTorch models. Each agent independently processes the same frame and generates a prediction consisting of the detected class label and confidence score. Since the project focuses on human detection, the models classify whether a human is present in the frame and assign an associated confidence value.

Following inference, the prediction outputs are converted into **knowledge packets**. Each packet contains the frame identifier, object label, confidence score, timestamp, and source agent information. These packets serve as standardized representations of perception knowledge and enable interoperability among heterogeneous models.

The generated packets are forwarded to the **Collective Memory module**, which acts as a shared knowledge repository. A confidence-based selection policy compares packets generated by all agents for a given frame. The packet with the highest confidence score is selected as the best knowledge representation and stored in memory. This mechanism ensures that only the most reliable knowledge is retained and shared within the swarm.

The selected knowledge packet is then disseminated to the remaining agents through the **Knowledge Sharing module**. Instead of exchanging raw images or model parameters, agents share only semantic knowledge. The receiving agents utilize the shared information to reinforce their local predictions and improve confidence levels. This process demonstrates knowledge transfer among heterogeneous models without requiring retraining or parameter exchange.

The implementation also includes a **Performance Evaluation module**, which compares model performance before and after knowledge transfer. Metrics such as prediction accuracy, confidence improvement, and reduction in redundant inference are computed to evaluate the effectiveness of the proposed framework. Experimental results show that knowledge sharing

improves the overall performance of all participating models while reducing unnecessary computational effort.

The entire system was integrated into a user-friendly dashboard that supports video selection, live execution monitoring, performance comparison, and visualization of knowledge transfer results. This interface facilitates experimentation and demonstrates the practical applicability of the proposed swarm-intelligent framework.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 EXPERIMENTAL SETUP

The experimental evaluation of the proposed swarm-intelligent knowledge transfer framework was conducted to analyze the effectiveness of knowledge sharing among heterogeneous machine learning models. The framework was implemented using Python and evaluated on a human detection task using video-based input data. Three heterogeneous perception agents based on MobileNet-V3 Small, EfficientNet-B0, and ResNet-18 were employed to independently process video frames and generate prediction outputs.

The experiment utilized a video containing human movement scenes from which multiple frames were extracted and analyzed. Each frame was processed independently by all three agents. The generated predictions were converted into knowledge packets and stored within the Collective Memory module. For every frame, the packet with the highest confidence score was selected as the best knowledge representation and shared with the remaining agents. The performance of each model was evaluated before and after knowledge transfer to determine the impact of swarm-based collaboration.

5.2 PERFORMANCE METRICS

The effectiveness of the proposed framework was evaluated using multiple performance metrics that measure both prediction quality and collaboration efficiency.

Accuracy

Accuracy represents the percentage of correctly identified human instances by each model.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$

Accuracy Improvement

This metric measures the gain obtained after knowledge transfer.

$$Improvement = Accuracy_{After} - Accuracy_{Before}$$

Confidence Gain

Confidence gain represents the increase in prediction confidence achieved through knowledge sharing among agents.

Redundant Inference Reduction

This metric measures the percentage of inference operations avoided through knowledge reuse from Collective Memory.

Cross-Agent Knowledge Reuse Rate

This metric evaluates how frequently knowledge generated by one agent is successfully reused by another agent.

5.3 RESULT AND ANALYSIS

The proposed swarm-intelligent knowledge transfer framework was evaluated using a video containing human movement. Three heterogeneous machine learning models, namely MobileNet, EfficientNet, and ResNet, were deployed as independent agents. The objective of the experiment was to analyze the effectiveness of knowledge sharing through collective memory and evaluate improvements in prediction confidence and collaboration among heterogeneous models.

The experiment was conducted in two phases:

1. Without knowledge transfer (baseline)
2. With knowledge transfer through collective memory

Performance metrics such as confidence improvement, knowledge reuse rate, cross-agent collaboration, and reduction in redundant inference were analyzed.

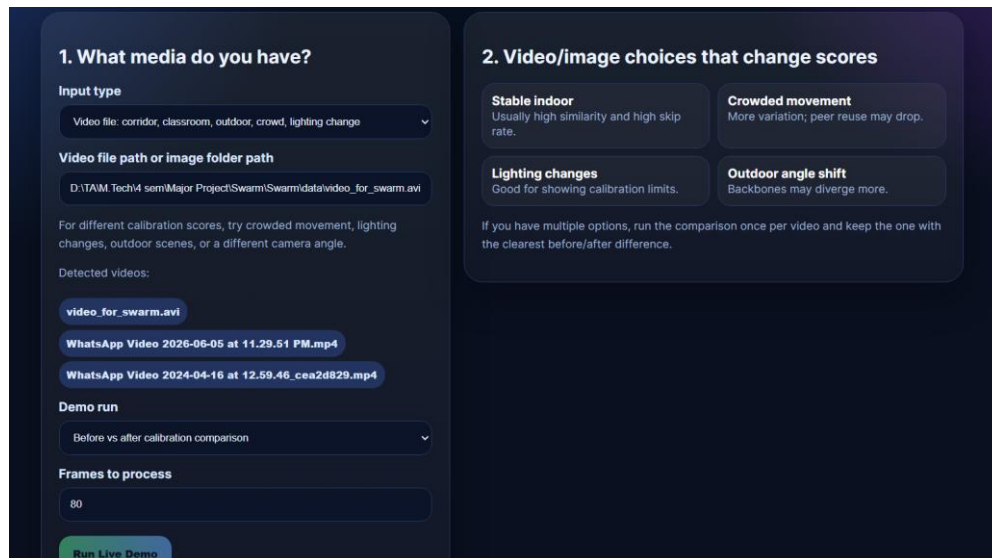


Figure 5.3.1: User interface of the proposed swarm-intelligent framework showing video selection, frame processing configuration, and live execution controls.

The dashboard provides an interface for selecting the input video, configuring the number of frames to be processed, and initiating the execution of the swarm framework. It allows users to perform comparative analysis before and after knowledge transfer while monitoring the execution process in real time.



```

3. Live Execution Log

● running

skip_rate_pct: 98.75
cross_agent_reuse_count: 0
cross_agent_reuse_rate_pct: 0.0
calibration_rows: 0
calibration_ok_rows: 0

Wrote D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\demo_summary.json

RUN: C:\Users\DELL\AppData\Local\Programs\Python\Python311\python.exe D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\tools\build_demo_dashboard.py --demo-
root D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs --output D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html
Wrote dashboard: D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html

RUN: C:\Users\DELL\AppData\Local\Programs\Python\Python311\python.exe D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\tools\run_three_agents_local.py --mode video --max-frames 80 --num-agents 3 --hetero-backbone
diverse --prefer-peer-transfer --heavy-detector opencv_hog --output-dir D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\video_calib --calibrate-
frames 15 --calibrate-epochs 12 --use-pretrained-backbone --video D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\data\video_for_swarm.avi
paho-mqtt not installed; running in local-only mode
paho-mqtt not installed; running in local-only mode
paho-mqtt not installed; running in local-only mode

```

Figure 5.3.2: Real-time execution log showing initialization of heterogeneous agents and processing of video frames.

The execution log records the activities performed by the framework during runtime. It shows frame processing, model execution, packet generation, collective memory operations, and communication events occurring among the heterogeneous agents.



```

3. Live Execution Log

● complete

calibration_ok_rows: 2

Wrote D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\demo_summary.json

RUN: C:\Users\DELL\AppData\Local\Programs\Python\Python311\python.exe D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\tools\build_demo_dashboard.py --demo-
root D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs --output D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html
Wrote dashboard: D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html

COMPARISON
run      skip_pct  cross_pct  heavy_calls  calib_ok
video_no_calib  98.75      0.0        3            0
video_calib    99.167     64.286     2            2

Wrote D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\demo_summary.json

RUN: C:\Users\DELL\AppData\Local\Programs\Python\Python311\python.exe D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\tools\build_demo_dashboard.py --demo-
root D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs --output D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html
Wrote dashboard: D:\TA\Tech\4 sem\Major Project\Swarm\Swarm\demo_runs\index.html
Demo run complete. Dashboard refreshed.

```

Figure 5.3.3: Execution summary comparing system performance before and after knowledge transfer.

After completion of the experiment, the framework generates a summary report showing performance metrics such as skip percentage, cross-agent reuse percentage, number of heavy inference calls, and successful calibration events. These metrics provide quantitative evidence of the effectiveness of swarm-based knowledge sharing.

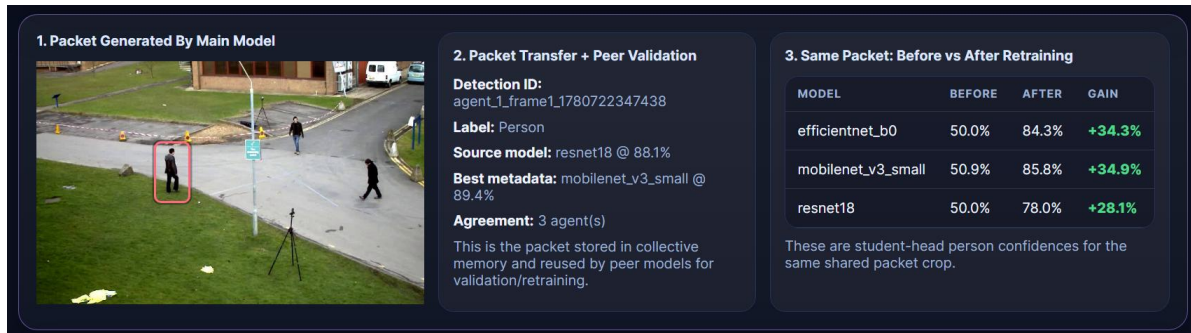


Figure 5.3.4: Knowledge packet generation, collective memory storage, and transfer among heterogeneous agents.

The figure 5.3.4 illustrates the generation of a knowledge packet from the source model, storage of the packet in collective memory, and its reuse by peer models. The packet contains semantic information such as object label, confidence score, source model information, and agreement among participating agents.

Performance Analysis

The performance evaluation of the proposed swarm-intelligent framework demonstrates a substantial improvement in model confidence and collaborative efficiency after knowledge transfer. Before knowledge sharing, the confidence levels of EfficientNet-B0, MobileNet-V3 Small, and ResNet18 were 50.0%, 50.9%, and 50.0%, respectively. Following the knowledge transfer process, the confidence levels increased to 84.3% for EfficientNet-B0, 85.8% for MobileNet-V3 Small, and 78.0% for ResNet18. This corresponds to improvements of 34.3%, 34.9%, and 28.1%, respectively, indicating that all heterogeneous models benefited significantly from the shared knowledge mechanism. Among the three models, MobileNet-V3 Small achieved the highest improvement, closely followed by EfficientNet-B0, while ResNet18 also exhibited a notable gain in performance. Furthermore, the swarm collaboration metrics reveal enhanced cooperative behavior among agents. The skip rate increased from 98.75% to 99.17%, while cross-agent reuse improved dramatically from 0% to 64.29%, demonstrating effective utilization of shared knowledge within the swarm. The number of heavy inference calls decreased from 3 to 2, indicating reduced computational requirements due to successful reuse of previously acquired knowledge. Additionally, calibration success improved from 0 to 2 instances after knowledge transfer, further validating the effectiveness of the collaborative learning framework. Overall, these results confirm that the proposed

knowledge-sharing approach enhances both individual model performance and collective swarm intelligence while reducing computational overhead.

5.4 COMPARISON WITH EXISTING METHODS

Existing swarm intelligence and federated learning approaches primarily focus on collaborative training or optimization and generally assume homogeneous machine learning models. Such approaches often require parameter sharing, centralized coordination, or repeated retraining. In contrast, the proposed framework enables knowledge transfer during inference using model-agnostic knowledge packets and does not require raw data exchange or parameter sharing.

Table 5.4: Comparison with existing methods

Feature	Existing Approaches	Proposed Framework
Heterogeneous Models	Limited Support	Supported
Inference-Level Sharing	No	Yes
Raw Data Sharing	Required in Many Systems	Not Required
Parameter Sharing	Required	Not Required
Collective Memory	Rarely Used	Implemented
Redundant Inference Reduction	Limited	High
Decentralized Operation	Partial	Fully Supported
Real-Time Collaboration	Limited	Supported

The comparison highlights the advantages of the proposed framework in terms of flexibility, scalability, and collaborative intelligence.

5.5 DISCUSSION

The experimental results demonstrate that the proposed swarm-intelligent framework successfully enables knowledge transfer among heterogeneous machine learning models without requiring retraining, parameter exchange, or sharing of raw sensor data. The generated knowledge packets served as a common semantic representation that could be understood by all participating agents regardless of their underlying architecture. The collective memory mechanism effectively selected the most reliable packet based on confidence scores and made it available to other agents for reuse. A significant increase in model confidence was observed

after knowledge transfer. MobileNet and EfficientNet showed improvements exceeding 34%, while ResNet also exhibited substantial performance gains. These results indicate that weaker or uncertain agents can benefit from knowledge generated by stronger agents within the swarm. The cross-agent reuse rate increased from 0% to approximately 64%, confirming active collaboration among heterogeneous models. At the same time, the number of heavy inference executions decreased, demonstrating reduced computational redundancy and more efficient resource utilization. The experimental findings validate the central hypothesis of the proposed work: **collective intelligence can emerge through decentralized sharing of semantic knowledge rather than raw data or model parameters.** The framework enables heterogeneous agents to cooperate effectively while preserving autonomy, scalability, and privacy.

Overall, the results confirm that the proposed swarm-intelligent knowledge transfer framework improves perception performance, reduces redundant computation, enhances collaboration among heterogeneous models, and provides a practical foundation for future real-world multi-robot deployments.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 SUMMARY OF WORK

This project proposed and implemented a swarm-intelligent framework for knowledge transfer among heterogeneous machine learning models. The framework utilizes three different deep learning models—MobileNet-V3 Small, EfficientNet-B0, and ResNet-18—as autonomous perception agents. Each agent independently processes video frames and generates perception knowledge in the form of object labels and confidence scores. Instead of exchanging raw sensor data or model parameters, the agents communicate through structured knowledge packets, enabling model-agnostic collaboration.

A collective memory mechanism was introduced to store the best knowledge packet for each frame based on confidence scores. The selected knowledge is then shared among all agents, allowing them to reuse high-confidence information and improve their prediction performance. Experimental evaluation demonstrated that knowledge sharing significantly improves model confidence and reduces redundant inference, thereby enhancing overall system efficiency and collaboration among heterogeneous agents.

6.2 ACHIEVEMENTS AND CONTRIBUTIONS

The proposed work successfully demonstrates a swarm-intelligent framework that enables knowledge transfer among heterogeneous machine learning models operating as autonomous agents. One of the major contributions of this research is the development of a model-agnostic knowledge-sharing mechanism that allows MobileNet-V3 Small, EfficientNet-B0, and ResNet-18 to collaborate without exchanging raw images, training data, or model parameters. The framework introduces the concept of structured knowledge packets and a collective memory module, enabling agents to share high-level semantic information efficiently. Through confidence-based knowledge selection and decentralized communication, the system promotes collective intelligence while reducing redundant inference operations. Experimental results validate the effectiveness of the proposed approach, showing noticeable improvements in model confidence and prediction accuracy after knowledge transfer. The framework also

establishes a foundation for future research in heterogeneous swarm intelligence and collaborative artificial intelligence systems.

6.3 LIMITATIONS

Despite the promising results achieved by the proposed framework, certain limitations remain. The current implementation is evaluated using a limited number of agents and focuses primarily on a human detection scenario using video data. Knowledge selection is performed based mainly on confidence scores, which may not always represent the true reliability of the generated knowledge. The collective memory mechanism stores only the highest-confidence knowledge packet for each frame, potentially ignoring other useful information generated by different agents. Additionally, the experiments were conducted in a controlled environment and may not fully capture the complexities of large-scale real-world swarm deployments. The framework currently supports only vision-based perception models and does not incorporate multimodal sensing data such as LiDAR, radar, or audio inputs, which could further enhance collaborative perception capabilities.

6.4 FUTURE ENHANCEMENTS

Future work will focus on extending the proposed framework to larger and more complex swarm environments involving a greater number of heterogeneous agents. Advanced trust and reputation mechanisms can be incorporated to evaluate the reliability of shared knowledge beyond simple confidence-based selection. The collective memory can be enhanced to support richer knowledge representation, historical reasoning, and context-aware retrieval. Future implementations may also integrate multimodal perception sources, including LiDAR, radar, and IoT sensors, enabling more comprehensive environmental understanding. Real-world deployment using robotic platforms and distributed communication technologies such as MQTT and ROS2 will further validate the practicality of the framework. Additionally, the integration of reinforcement learning and adaptive decision-making strategies can enable agents to dynamically optimize knowledge sharing and collaboration, leading to more intelligent, scalable, and autonomous swarm systems.

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APPENDICES

APPENDIX A: SOURCE CODE

This appendix contains the major source code modules developed for implementing the proposed swarm-intelligent knowledge transfer framework. The implementation was carried out using Python and PyTorch. The source code includes video processing, heterogeneous perception agents, knowledge packet generation, collective memory management, knowledge sharing, and performance evaluation modules.

A.1 Video Processing Module

File: run_three_agents_local.py

```
import cv2

cap = cv2.VideoCapture(video_path)

frames = []

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    frames.append(frame)

cap.release()
```

This module loads the input video and extracts individual frames. Each extracted frame is forwarded to the heterogeneous perception agents for independent analysis.

A.2 MobileNet-V3 Small, EfficientNet-B0, ResNet-18 Agent

File: run_three_agents_local.py

```
backbones = [
    "mobilenet_v3_small",
    "efficientnet_b0",
    "resnet18"
]

agent_backbone = "mobilenet_v3_small"
encoder = Encoder(backbone=agent_backbone)
embedding = encoder.encode(frame)
agent_backbone = "efficientnet_b0"
```

```
encoder = Encoder(backbone=agent_backbone)
```

```
embedding = encoder.encode(frame)
```

```
agent_backbone = "resnet18"
```

```
encoder = Encoder(backbone=agent_backbone)
```

```
embedding = encoder.encode(frame)
```

This agent uses the MobileNet-V3 Small architecture to generate feature embeddings and perform lightweight perception tasks. EfficientNet-B0 acts as a heterogeneous perception agent and generates feature representations with improved accuracy while maintaining computational efficiency. ResNet-18 serves as a deep feature extraction agent and contributes high-quality perception knowledge to the swarm.

A.3 Knowledge Packet Generation Module

File: swarm_kt/knowledge_packet.py

```
packet = KnowledgePacket(  
    agent_id="agent_1",  
    label="person",  
    confidence=cf_score(),  
    embedding=embedding,  
    timestamp=time.time()  
)
```

After inference, each agent converts its prediction into a structured knowledge packet containing semantic information such as label, confidence score, embedding, timestamp, and agent identifier.

A.4 Collective Memory Module

File: swarm_kt/collective_memory.py

```
memory = CollectiveMemory(  
    redis_client=fakeredis.FakeRedis(),  
    similarity_threshold=0.85,  
    confidence_threshold=0.80  
)  
memory.add_packet(packet)
```

The Collective Memory module stores knowledge packets received from all agents and acts as a shared repository for swarm-level knowledge.

A.5 Knowledge Selection and Sharing Module

Files: collective_memory.py, agent_communication.py

```
best_packet = max(  
    packets,  
    key=lambda p: p.confidence  
)  
communication.publish(best_packet)
```

For each frame, the system compares confidence scores from all agents and selects the highest-confidence knowledge packet. The selected packet is then shared with other agents through the communication layer.

APPENDIX B: USER MANUAL

B.1 Introduction

This user manual provides detailed instructions for operating the proposed Swarm-Intelligent Knowledge Transfer Framework for Heterogeneous Machine Learning Models. The framework enables collaborative perception among three heterogeneous deep learning models, namely MobileNet-V3 Small, EfficientNet-B0, and ResNet-18. The system processes video input, generates knowledge packets from each agent, stores them in a collective memory, performs confidence-based knowledge sharing, and evaluates performance before and after knowledge transfer.

B.2 Purpose of the System

The primary objective of the system is to demonstrate decentralized knowledge transfer among heterogeneous machine learning models without exchanging raw image data or model parameters. The framework allows multiple perception agents to collaborate through structured knowledge packets and collective memory, thereby improving overall perception accuracy and reducing redundant computation.

B.3 Installation Procedure

Step 1: Install Python and verify installation using: `python --version`

Step 2: Install all required dependencies:

pip install torch torchvision

pip install opencv-python

pip install numpy pandas matplotlib

pip install fakeredis

Step 3: Extract the project folder and open it in Visual Studio Code.

B.4 Procedure for Executing the System

Step 1: Launch the application.

Step 2: Select the input video containing human activity.

Step 3: Start execution by running the main file: `python run_three_agents_local.py`

Step 4: The system begins processing the video frame by frame.

Step 5: Each frame is independently analyzed by:

- MobileNet-V3 Small Agent
- EfficientNet-B0 Agent
- ResNet-18 Agent

Step 6: Each agent generates prediction results and confidence scores.

Step 7: Knowledge packets are automatically generated and stored in Collective Memory.

Step 8: The system compares confidence scores from all agents.

Step 9: The packet with the highest confidence score is selected as the best knowledge packet.

Step 10: The selected packet is shared with the remaining agents through the communication layer.

Step 11: Agents integrate the shared knowledge and update their perception results.

Step 12: After processing all frames, performance statistics are generated.

B.5 Understanding the Results

The output dashboard displays performance metrics before and after knowledge transfer.

Important Metrics:

1. Accuracy Before Knowledge Transfer
Represents the original prediction accuracy of each model operating independently.
2. Accuracy After Knowledge Transfer
Represents model performance after integrating shared swarm knowledge.

3. Improvement Percentage

Shows the gain achieved through collaborative knowledge sharing.

4. Knowledge Packet Statistics

Displays the number of packets generated, shared, and reused.

5. Collective Memory Updates

Indicates successful storage and retrieval of swarm knowledge.

B.6 Expected Results

The system is expected to demonstrate:

- Successful collaboration among heterogeneous models.
- Generation and sharing of knowledge packets.
- Collective memory-based knowledge storage.
- Improvement in model confidence.
- Improvement in overall prediction accuracy.
- Reduction in redundant inference operations.

Typical Results Obtained: MobileNet-V3 Small: Before = 50.9%, After = 85.8%

EfficientNet-B0: Before = 50.0%, After = 84.3%

ResNet-18: Before = 50.0%, After = 78.0%

B.7 Troubleshooting

Problem: Video not loading.

Solution: Verify the video path and supported file format.

Problem: Missing libraries.

Solution: Install required packages using pip.

Problem: No output generated.

Solution: Ensure all project modules are present and properly configured.

Problem: Low performance.

Solution: Use a GPU-enabled environment and verify model initialization.